

Teo convergencia serie imp lim 0

Teorema 1. Sea $\sum_n a_n$ una serie convergente $\implies \lim_n a_n = 0$.

Demostración: Sea $(s_n)_{n \in \mathbb{N}}$ dada por $s_n = \sum_{k=0}^n a_k$, entonces s_n converge a algún s luego $(s_n)_{n \in \mathbb{N}}$ es de Cauchy. Es decir, $\forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall n, m \geq N_0 : |s_n - s_m| < \varepsilon$.

Tomando $m = n - 1$, $|s_n - s_{n-1}| = |a_n| < \varepsilon$. Por tanto, $\lim_{n \rightarrow \infty} a_n = 0$. ■

Referenciado en

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